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Sec: U

Course: Compiler Design

Task 1:

Write a C++ program that takes a string as input (a candidate variable name) and prints whether it is **Valid** or **Invalid** using these rules:

1. Invalid if it starts with a digit
 2. Invalid if it matches a reserved keyword, check against **any 6 to 8 common C++ keywords**
 3. Invalid if it contains special characters other than underscore _ or dollar sign \$
- You may use string.at() to inspect characters.

Sample Input:

1var

Expected Output:

Invalid variable name

Answer:

```
#include <iostream>

#include <string>

#include <cctype>

using namespace std;

int main() {

    string varName;

    cout << "Enter variable name: ";

    cin >> varName;

    string keywords[] = {"int", "float", "double", "char", "if", "else", "for", "while"};

    if (isdigit(varName[0])) {

        cout << "Invalid variable name" << endl;

        return 0;}

    for (string keyword : keywords) {

        if (varName == keyword) {

            cout << "Invalid variable name" << endl;

            return 0;

        }

    }
```

```

    }

    for (char ch : varName){

        if (!isalnum(ch) && ch != '_' && ch != '$'){

            cout << "Invalid variable name" << endl;

            return 0;

        }

    }

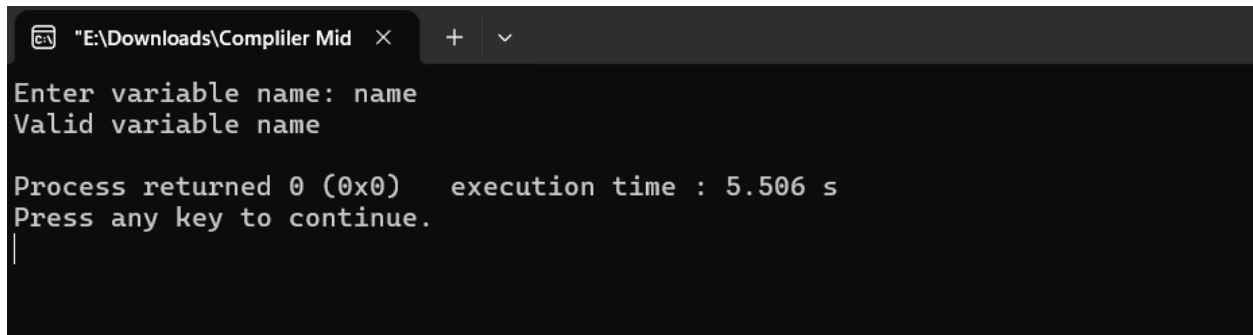
    cout << "Valid variable name" << endl;

    return 0;

}

```

Output:

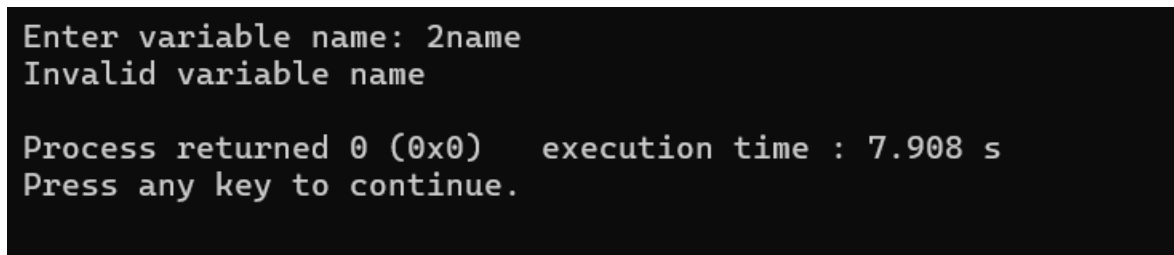


```

E:\Downloads\Compiler Mid  ×  +  ▾
Enter variable name: name
Valid variable name

Process returned 0 (0x0)   execution time : 5.506 s
Press any key to continue.
|

```



```

Enter variable name: 2name
Invalid variable name

Process returned 0 (0x0)   execution time : 7.908 s
Press any key to continue.

```

Task 2:

Find the **average value** of the elements of an array. You do **not** need to take the array as input, define it in the program.

Sample (Fixed Array in Code):

```
int a[] = {10, 20, 30, 40, 50};
```

Expected Output:

Average value = 30

Answer:

```
#include <iostream>

using namespace std;

int main() {

    int a[] = {10, 20, 30, 40, 50};

    int size = 5;

    int sum = 0;

    for (int i = 0; i < size; i++) {

        sum = sum + a[i];

    }

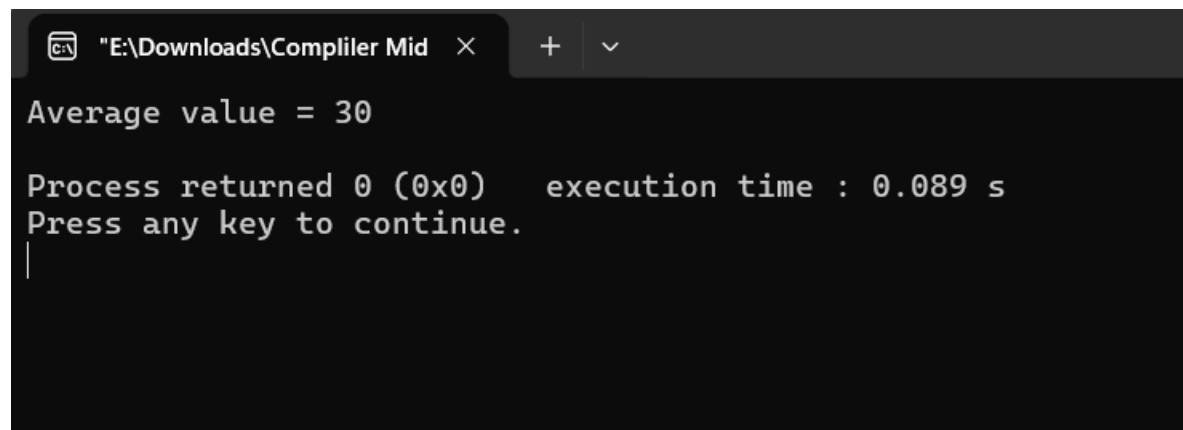
    float average = sum / size;

    cout << "Average value = " << average << endl;

    return 0;

}
```

Output:

A screenshot of a C++ compiler window titled "E:\Downloads\Compiler Mid". The window shows the output of the program: "Average value = 30". Below this, it displays "Process returned 0 (0x0) execution time : 0.089 s" and "Press any key to continue." with a cursor on the line below.

```
"E:\Downloads\Compiler Mid" × + ∨

Average value = 30

Process returned 0 (0x0) execution time : 0.089 s
Press any key to continue.
|
```

Task 3:

Write a C++ program that takes a single token (no spaces) and prints:

- Keyword if it matches one of these keywords: int, float, double, char, if, else, while, return
- Identifier if it is a valid identifier (starts with letter or underscore, then letters, digits, underscores only)
- Invalid otherwise

Sample Input:

while

Expected Output:

Keyword

Answer:

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int a[] = {7, -2, 15, 4, 9};
```

```
    int size = 5;
```

```
    int min = a[0];
```

```
    int max = a[0];
```

```
    for (int i = 1; i < size; i++) {
```

```
        if (a[i] < min) {
```

```
            min = a[i];
```

```
        }
```

```
        if (a[i] > max) {
```

```
            max = a[i];
```

```
        }
```

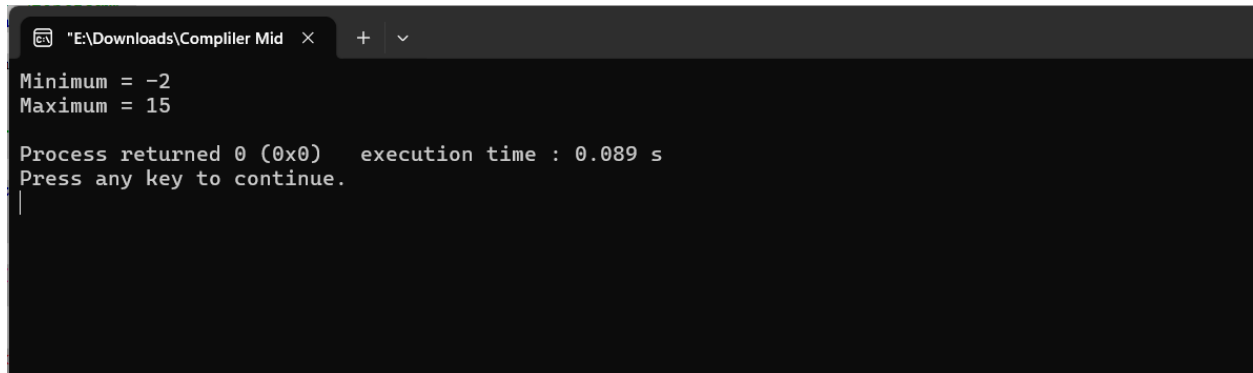
```
    }
```

```
    cout << "Minimum = " << min << endl;
```

```
    cout << "Maximum = " << max << endl;
```

```
    return 0;
}
```

Output:



```
"E:\Downloads\Compiler Mid"
Minimum = -2
Maximum = 15

Process returned 0 (0x0)   execution time : 0.089 s
Press any key to continue.
```

Task 4:

Write a C++ program that takes a single token and prints:

- Integer literal if it contains only digits, optionally preceded by + or -
- Floating literal if it contains digits with exactly one decimal point, optionally preceded by + or -, and at least one digit on both sides of the decimal point
- Invalid number otherwise

Sample Input:

-12.50

Expected Output:

Floating literal

Answer:

```
#include <iostream>
```

```
#include <string>
```

```
using namespace std;
```

```
int main() {
```

```
    string firstName, lastName;
```

```
    cout << "First name: ";
```

```
    cin >> firstName;
```

```

    cout << "Last name: ";

    cin >> lastName;

    string fullName = firstName + " " + lastName;

    cout << "Full name = " << fullName << endl;

    return 0;
}

```

Output:

```

First name: Rahin
Last name: Ibne Harun
Full name = Rahin Ibne

Process returned 0 (0x0)   execution time : 8.719 s
Press any key to continue.
|

```

Task 5:

Write a C++ program that takes a full line as input, prints each delimiter in order, then prints the total count. Use these delimiters:

(){}[],;

Sample Input:

```
int f(int a, int b){ return a+b; }
```

Expected Output:

delimiter1 : (

delimiter2 : (

delimiter3 : ,

delimiter4 :)

delimiter5 :)

```
delimiter6 : {  
delimiter7 : ;  
delimiter8 : }  
  
number of delimiters = 8
```

Answer:

```
#include <iostream>  
  
#include <fstream>  
  
#include <string>  
  
using namespace std;  
  
int main() {  
    int n;  
  
    cout << "Number of students: ";  
  
    cin >> n;  
  
    ofstream outFile("students.txt");  
  
    for (int i = 0; i < n; i++) {  
        string name;  
  
        int marks;  
  
        cout << "Student " << i + 1 << " name: ";  
  
        cin >> name;  
  
        cout << "Student " << i + 1 << " marks: ";  
  
        cin >> marks;  
  
        outFile << name << " " << marks << endl;  
    }  
  
    outFile.close();  
  
    cout << "\nData read from file:" << endl;
```



```
ifstream inFile("students.txt");

string name;

int marks;

while (inFile >> name >> marks) {

    cout << name << " " << marks << endl;

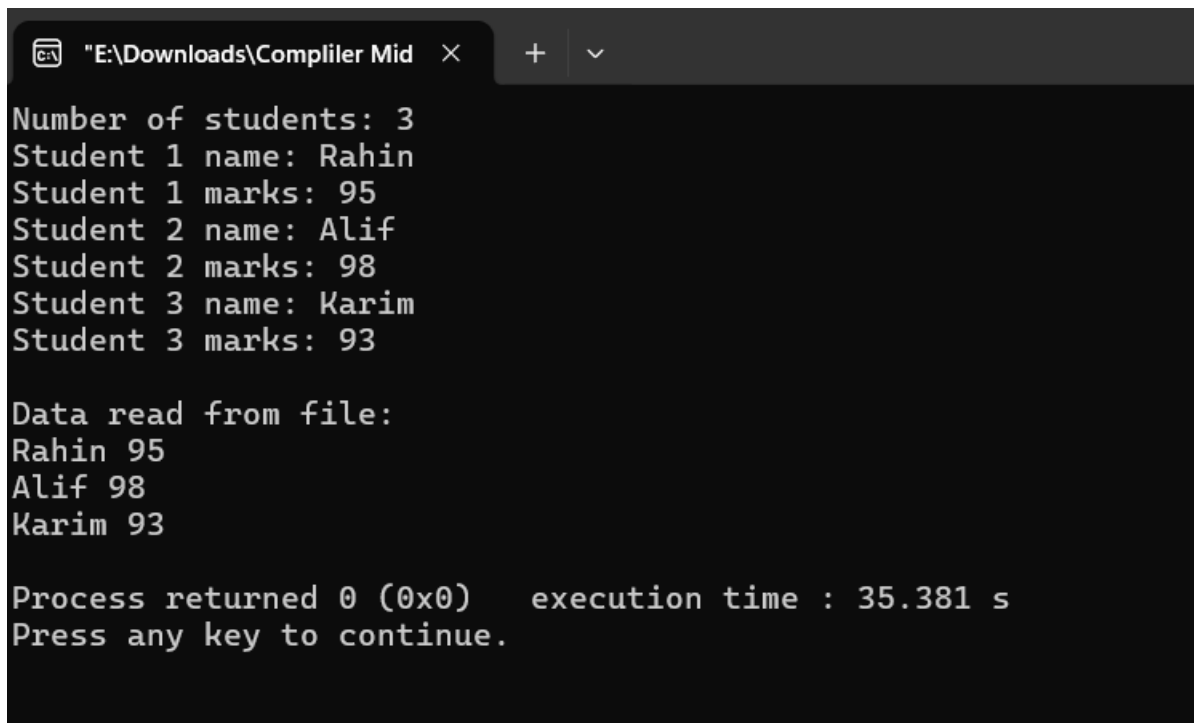
}

inFile.close();

return 0;

}
```

Output:



```
"E:\Downloads\Compiler Mid" × + v

Number of students: 3
Student 1 name: Rahin
Student 1 marks: 95
Student 2 name: Alif
Student 2 marks: 98
Student 3 name: Karim
Student 3 marks: 93

Data read from file:
Rahin 95
Alif 98
Karim 93

Process returned 0 (0x0)    execution time : 35.381 s
Press any key to continue.
```

//////////////////// **End**////////////////////